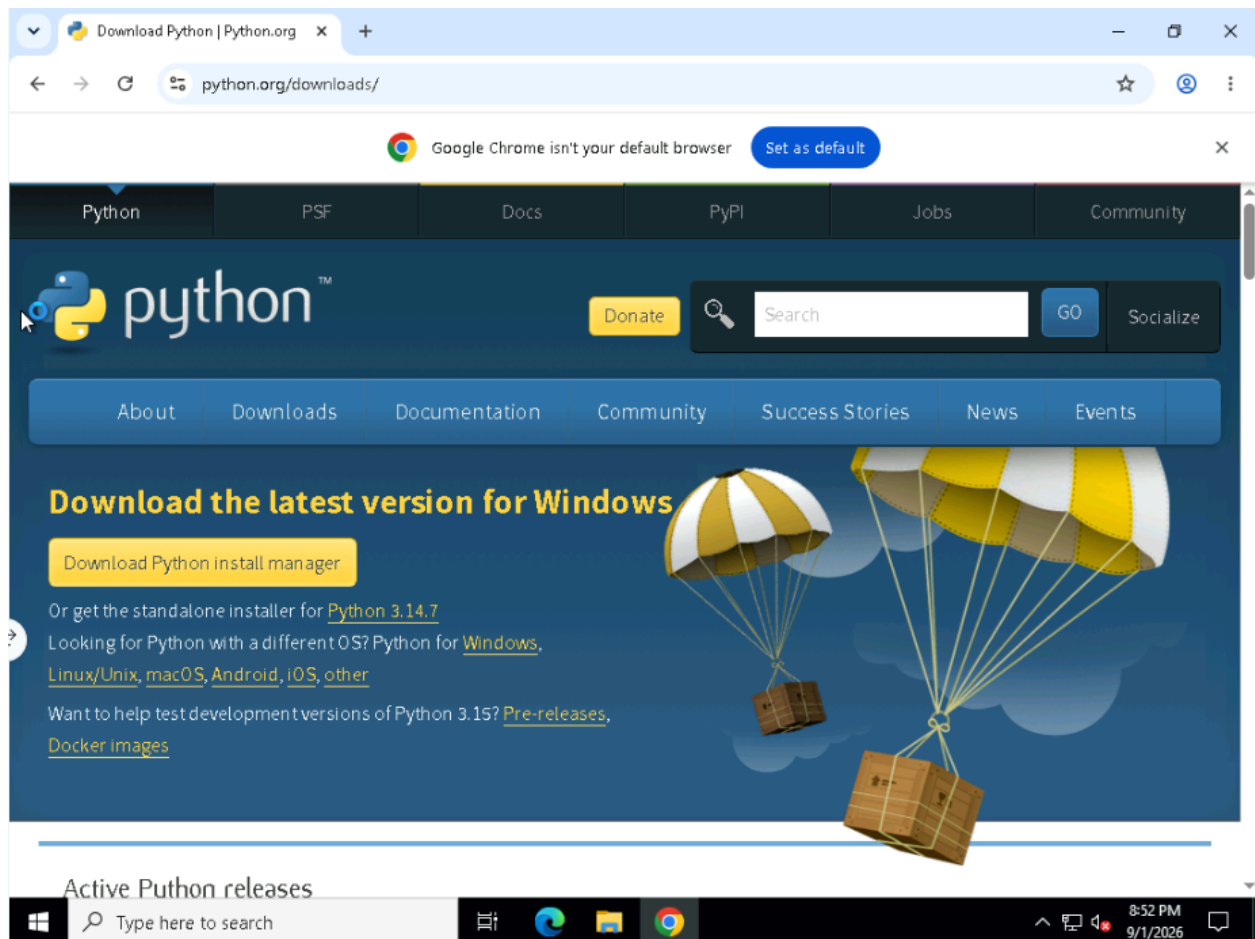
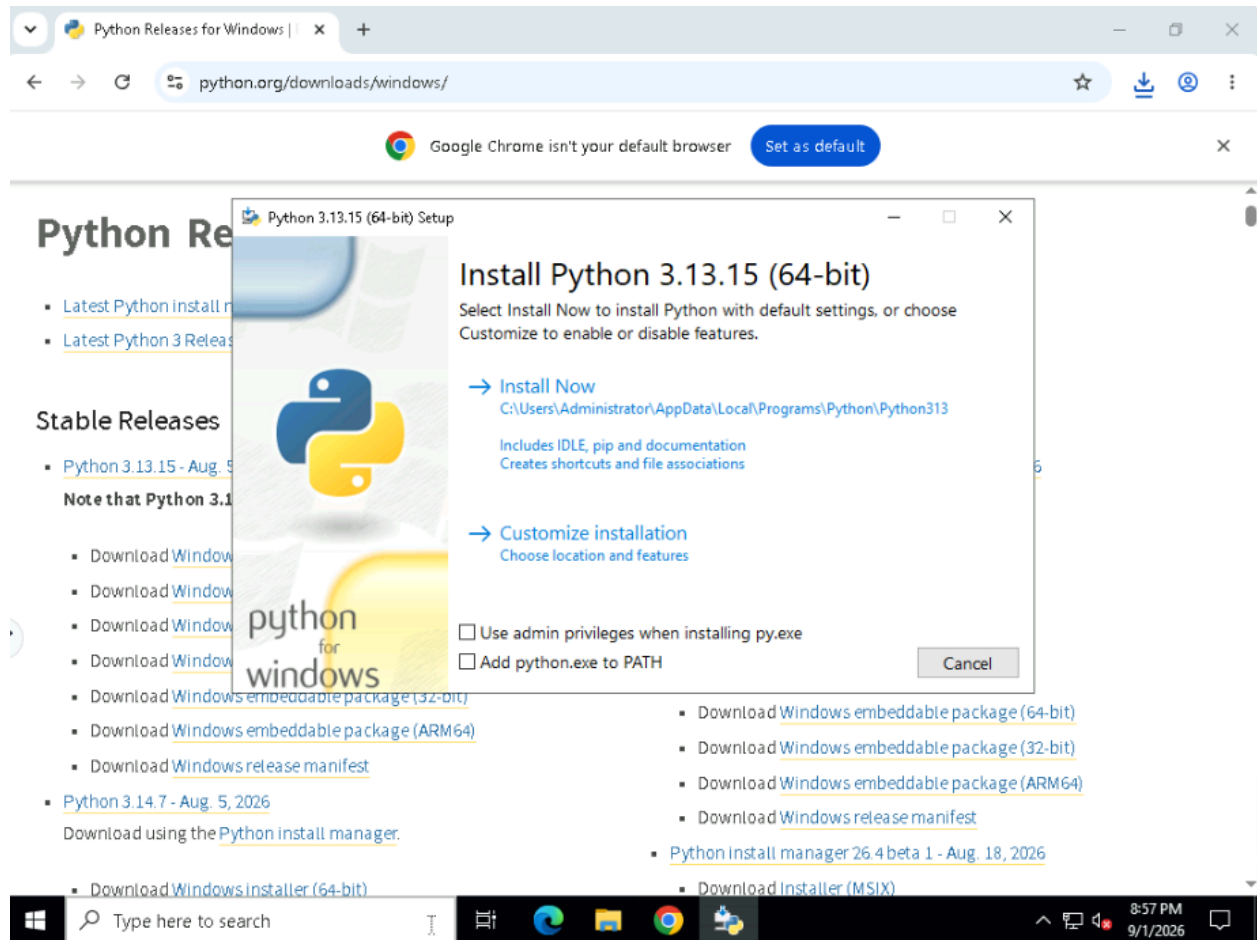


Task1.) Install the Latest Version of Python

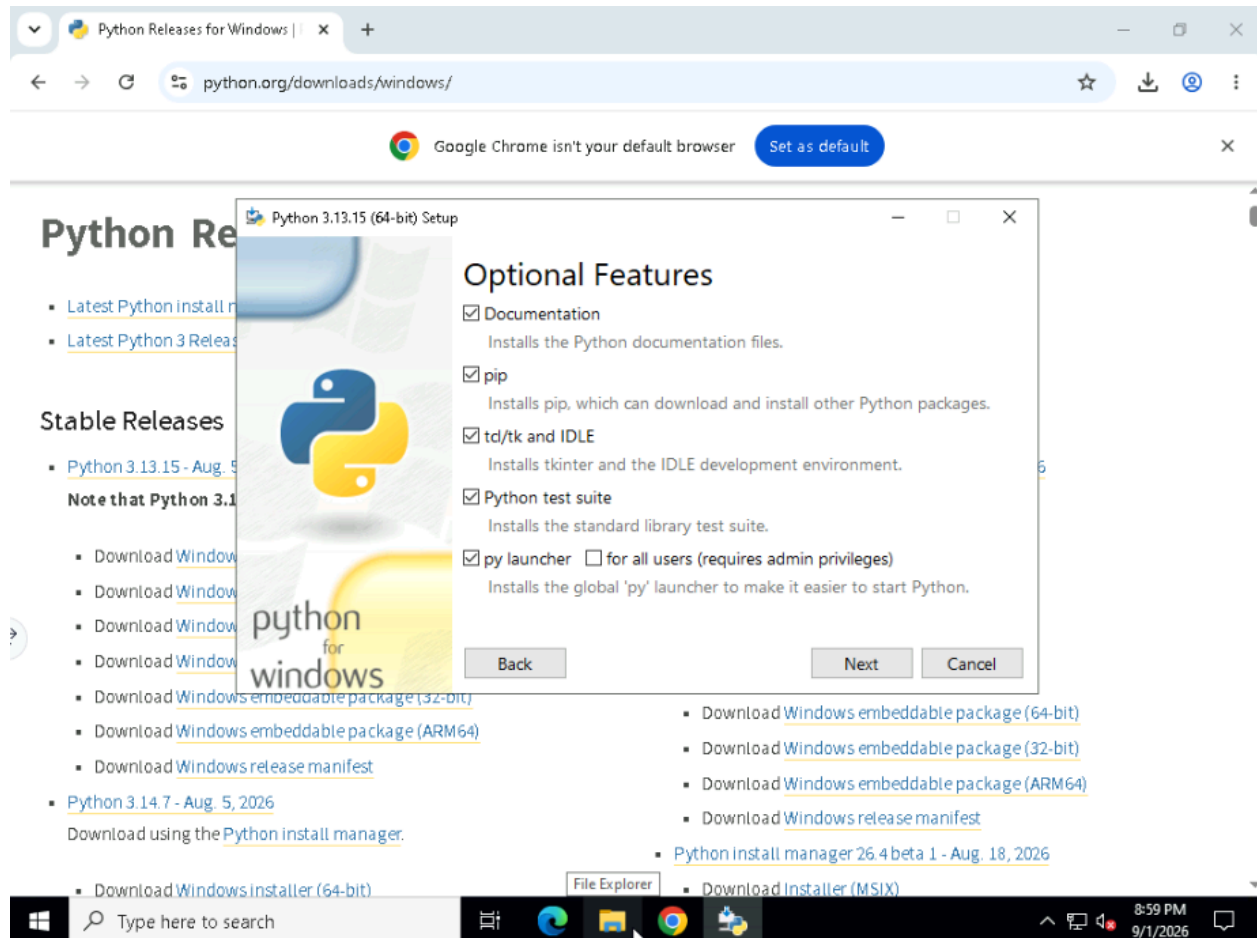
1.) Go to <https://www.python.org/downloads/>.



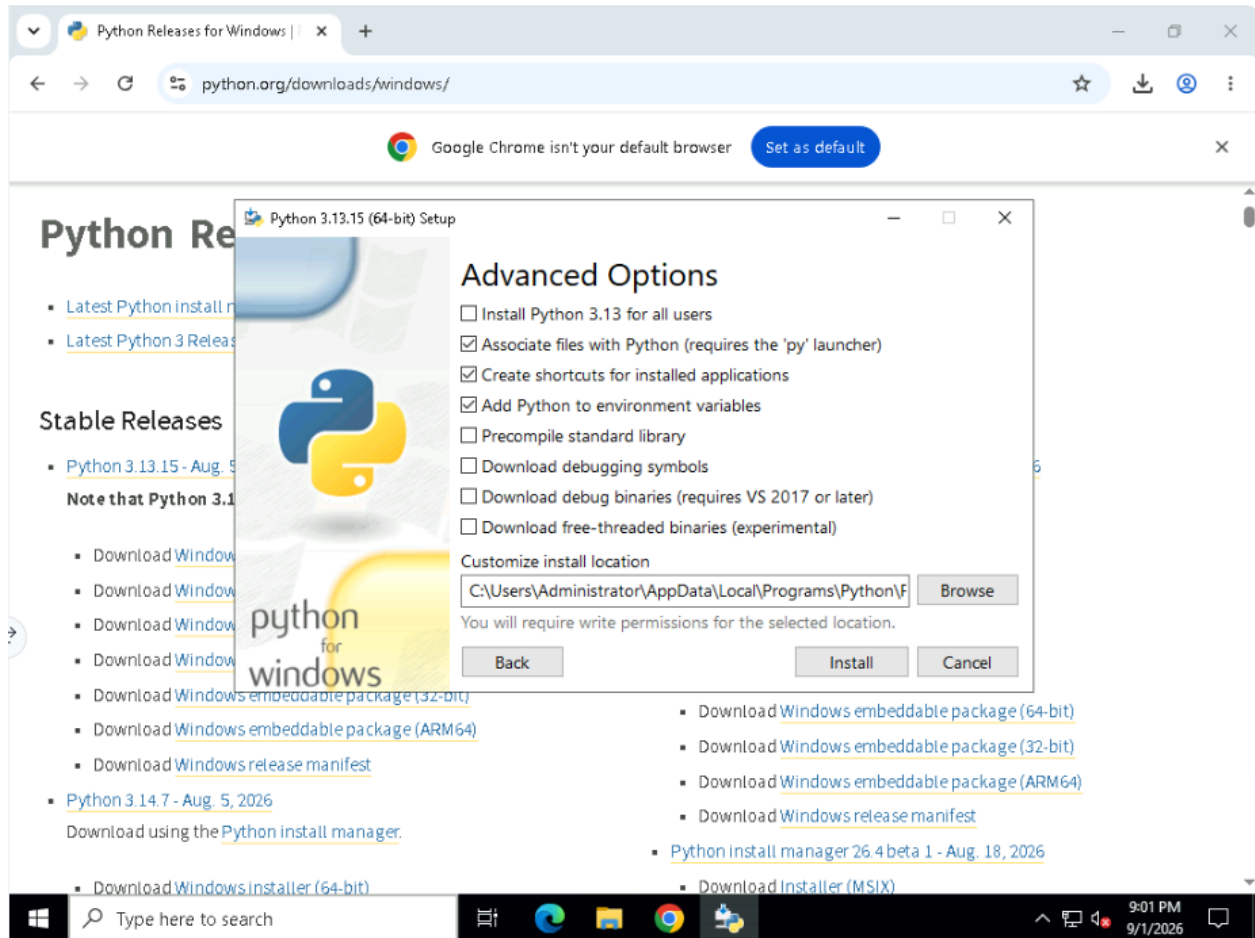
2.) Select Customize installation.



- 3.) On the Optional Features window, select the following options and click Next.
- a.) Documentation
 - b.) Pip
 - c.) tcl/tk and IDLE
 - d.) Python test suite
 - e.) Py launcher



- 4.) On the Advanced Options window, select the following options and click Install
- Associate files with Python (requires the 'py' launcher)
 - Create shortcuts for installed applications
 - Add Python to environment variables



Task 2.) Install Scapy

- 1.) You'll need Scapy library to use the Sorst Scan Detector Python script. Scapy is a powerful Python library used for network packet manipulation and sending and receiving network packets.
 - a.) To install Scapy, open command prompt and type "pip install scapy"

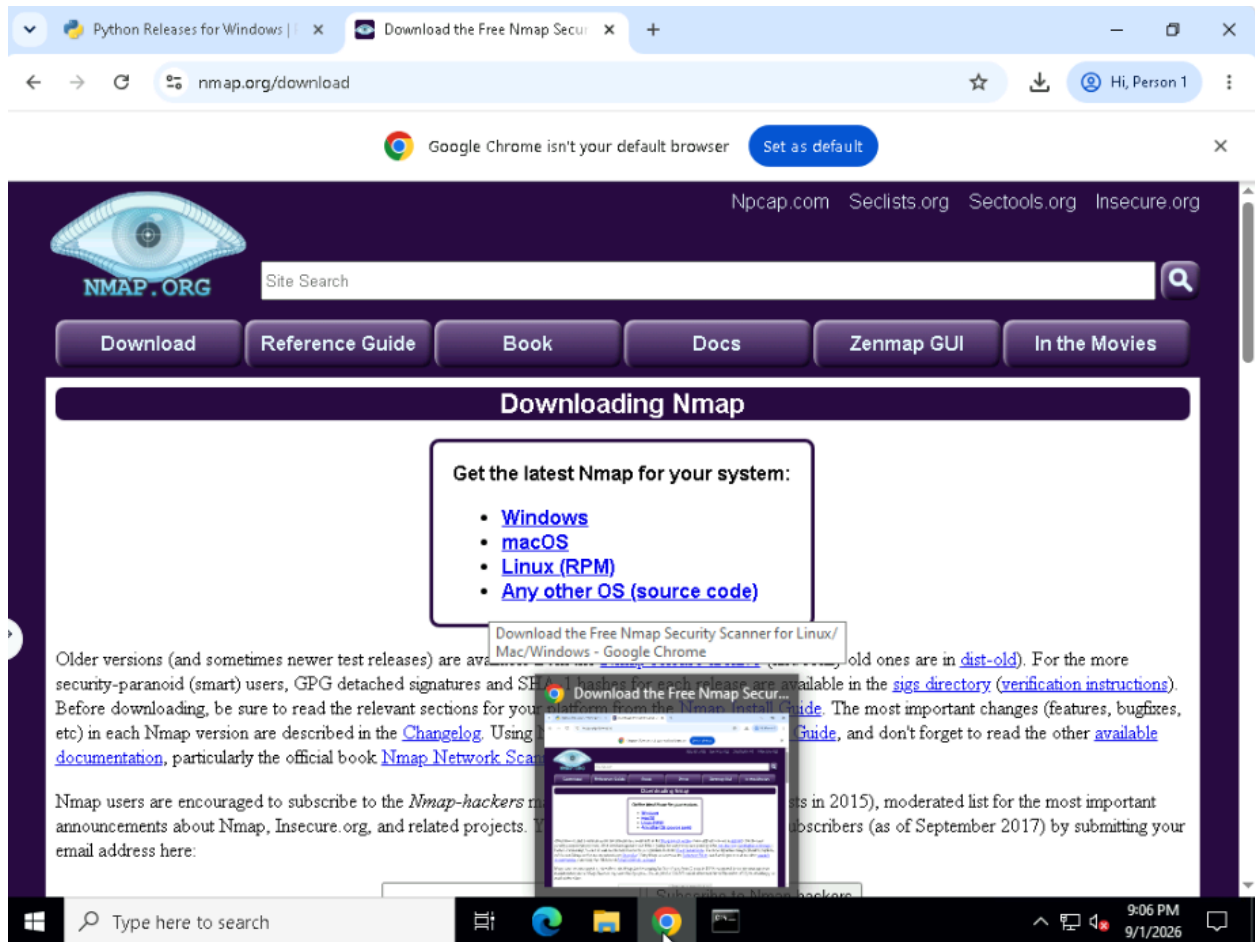
```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.4297]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>pip install scapy
Collecting scapy
  Downloading scapy-2.7.0-py3-none-any.whl.metadata (5.8 kB)
  Downloading scapy-2.7.0-py3-none-any.whl (2.6 MB)
----- 2.6/2.6 MB 46.3 MB/s 0:00:00
Installing collected packages: scapy
Successfully installed scapy-2.7.0

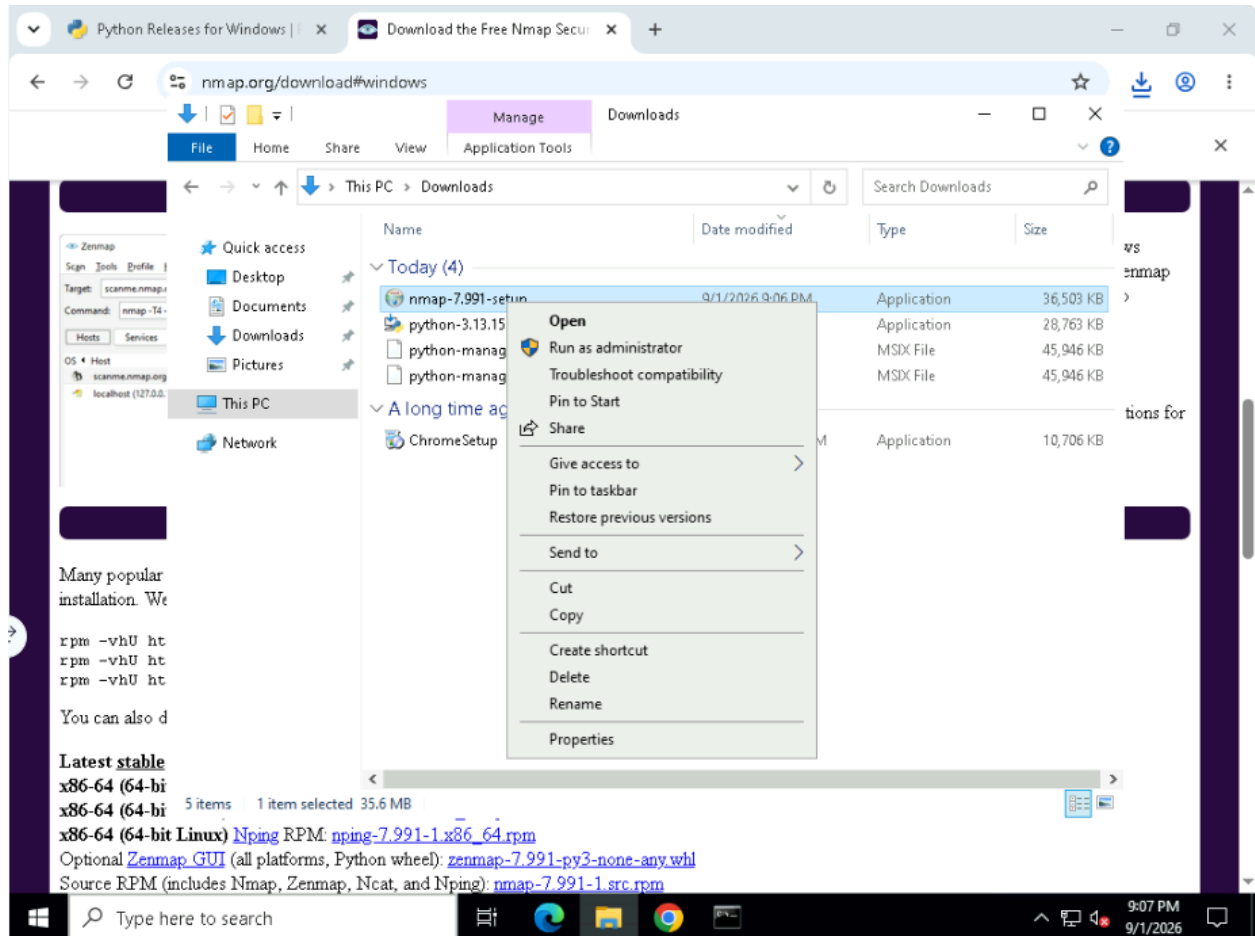
C:\Users\Administrator>
```

Task 3.) Install nmap for Port Scanning

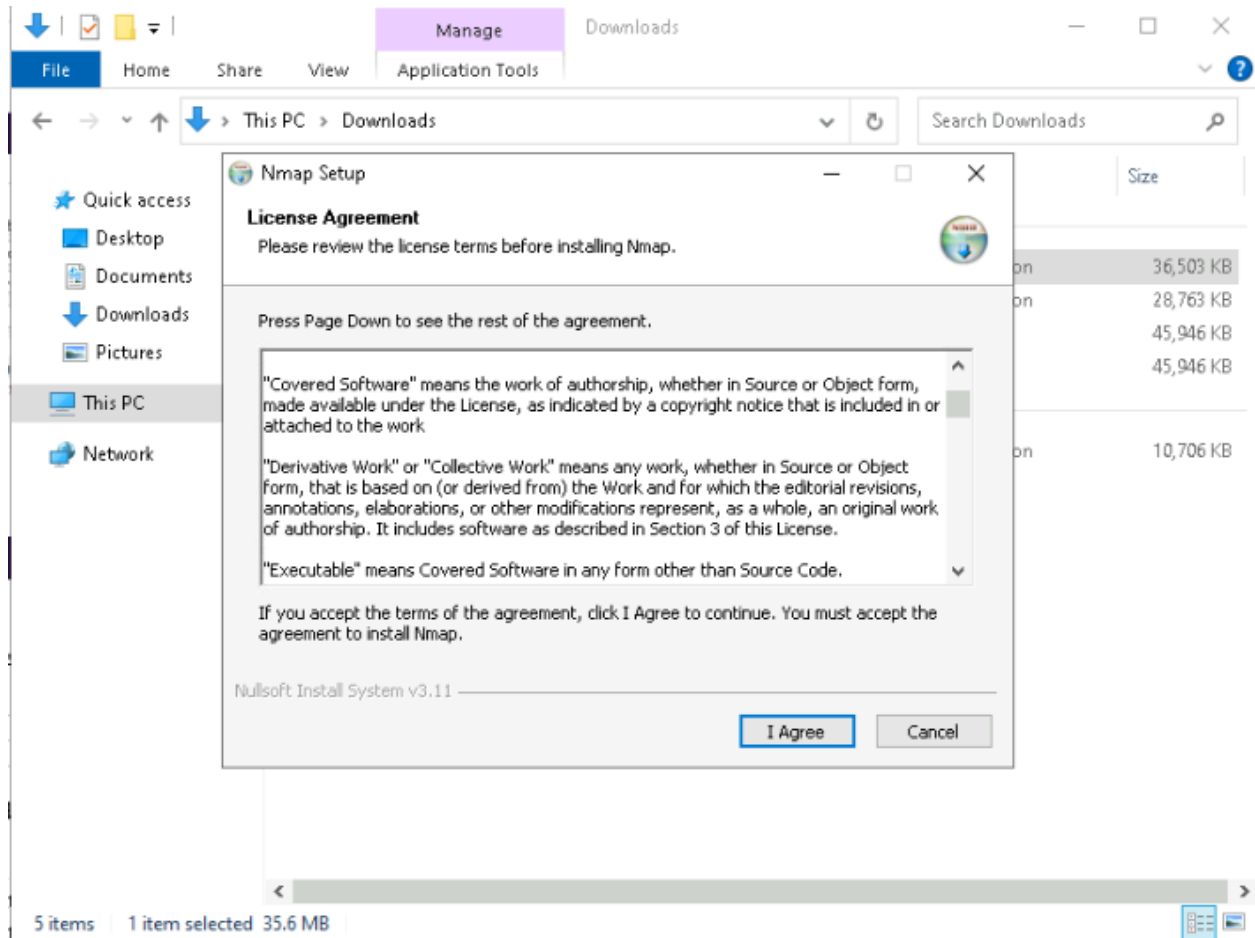
- 1.) Download nmap from <https://nmap.org/download>



- 2.) Verify that you have administrator privileges for the computer
 - a.) Navigate the folder where you download nmap
 - b.) Right click the file name and install the application using administrative privileges

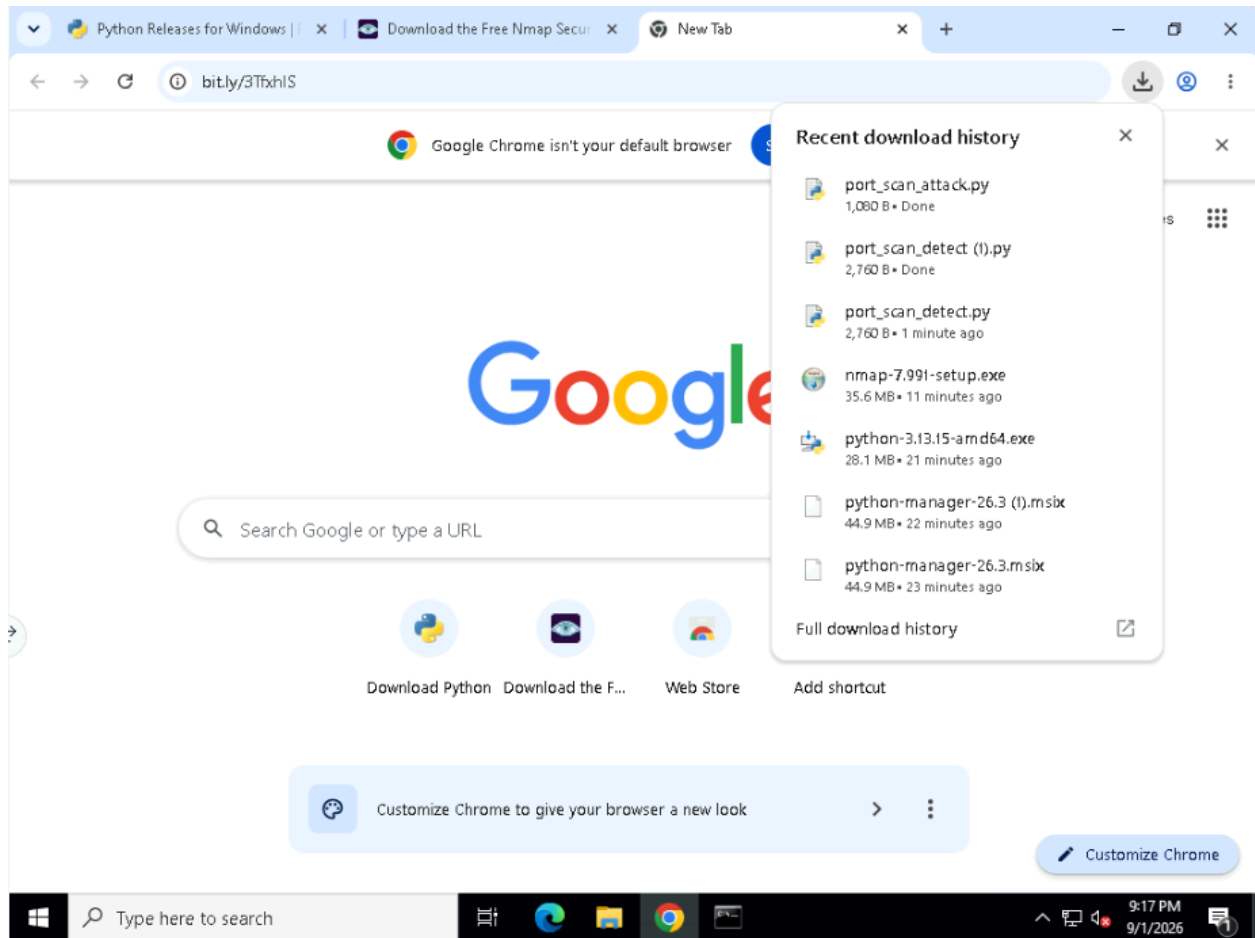


3.) Install the application with all default options selected

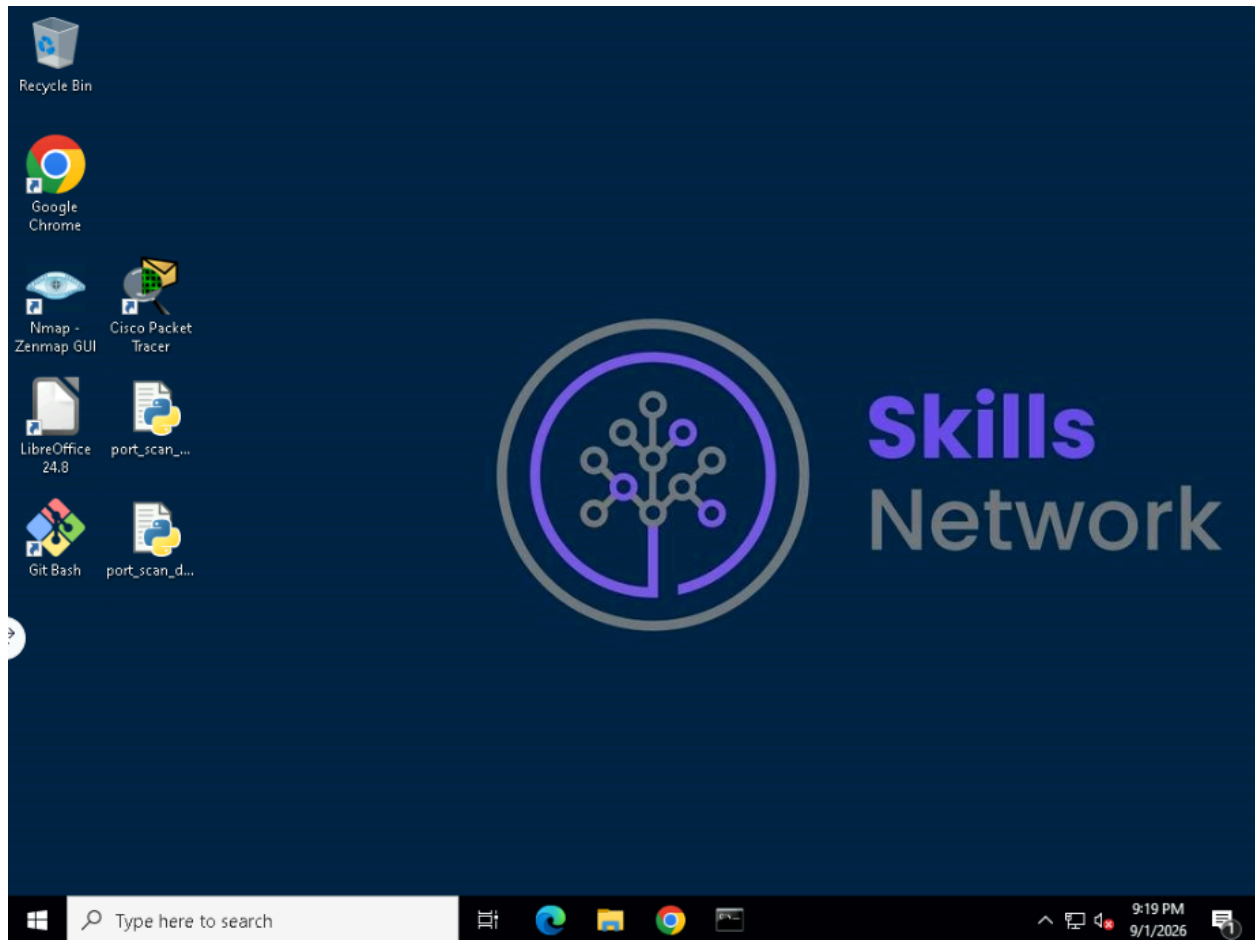


Task 4.) Download the Port Scan Attacker and Detector

- 1.) Download the Port Scan Detect file at <https://bit.ly/4ed1Bw3>
 - a.) A file named port_scan_detect.py should be downloaded
- 2.) Download the Port Scan Attack file at <https://bit.ly/3TfxhIS>
 - a.) A file named port_scan_attack.py should be downloaded



3.) Make sure the files are available on the desktop

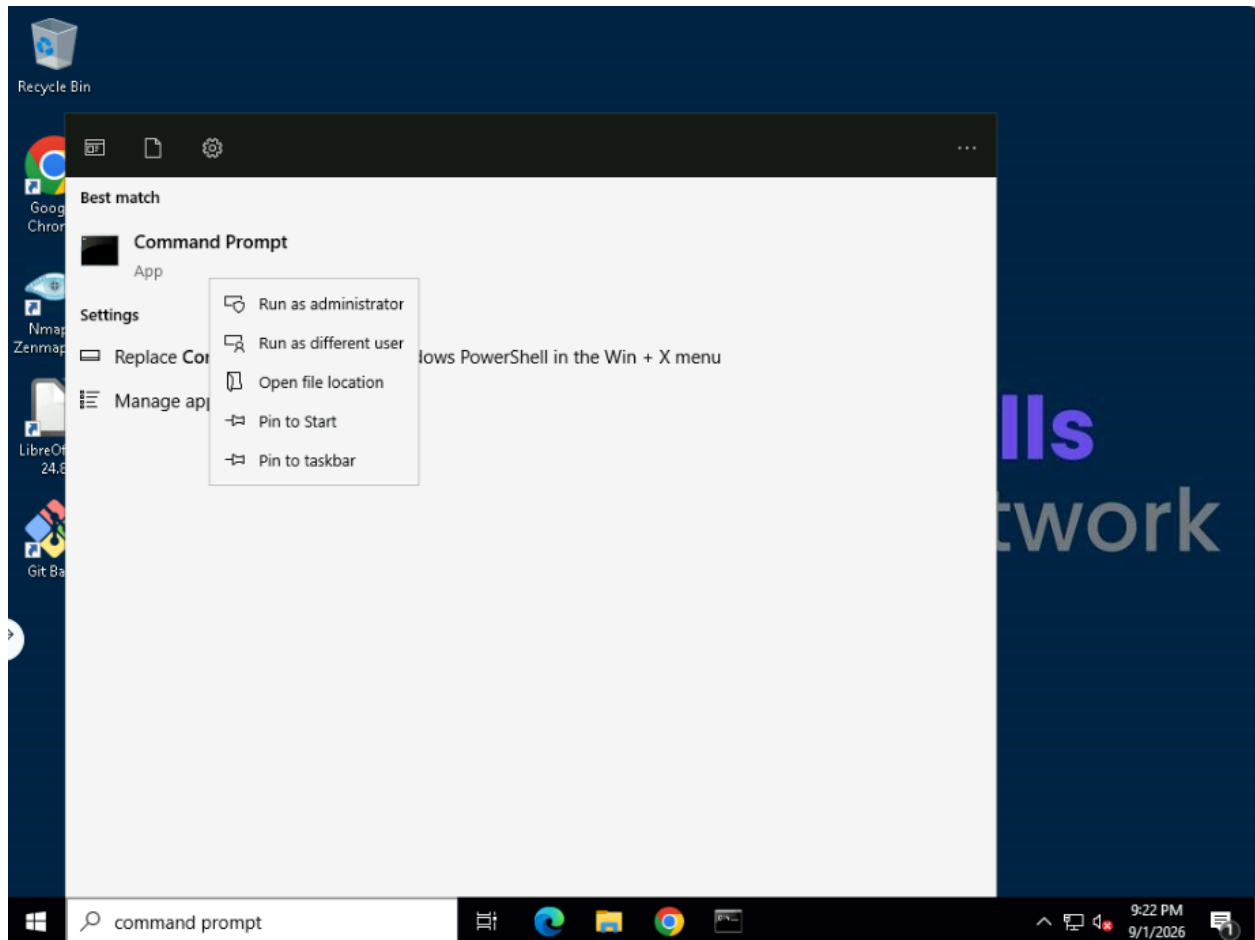


Task 5.) Port Scanning and Detection

Scenario: After mitigating a Ping of Death attack, you notice unusual network traffic patterns. Suspecting that an attacker might be scanning your network for open ports, you decide to perform a port scan to understand how an attacker might exploit your network. After performing this process, you will verify if your intrusion detection system can detect such scanning activities.

1.) Perform a port scan:

- a.) Search command prompt and right click on it and select Run as Administrator



- 2.) In command prompt, navigate to the folder where port_scan_attack.py is located
 - a.) Type “cd desktop”
 - b.) Type “python port_scan_attack.py” to run the port scan script

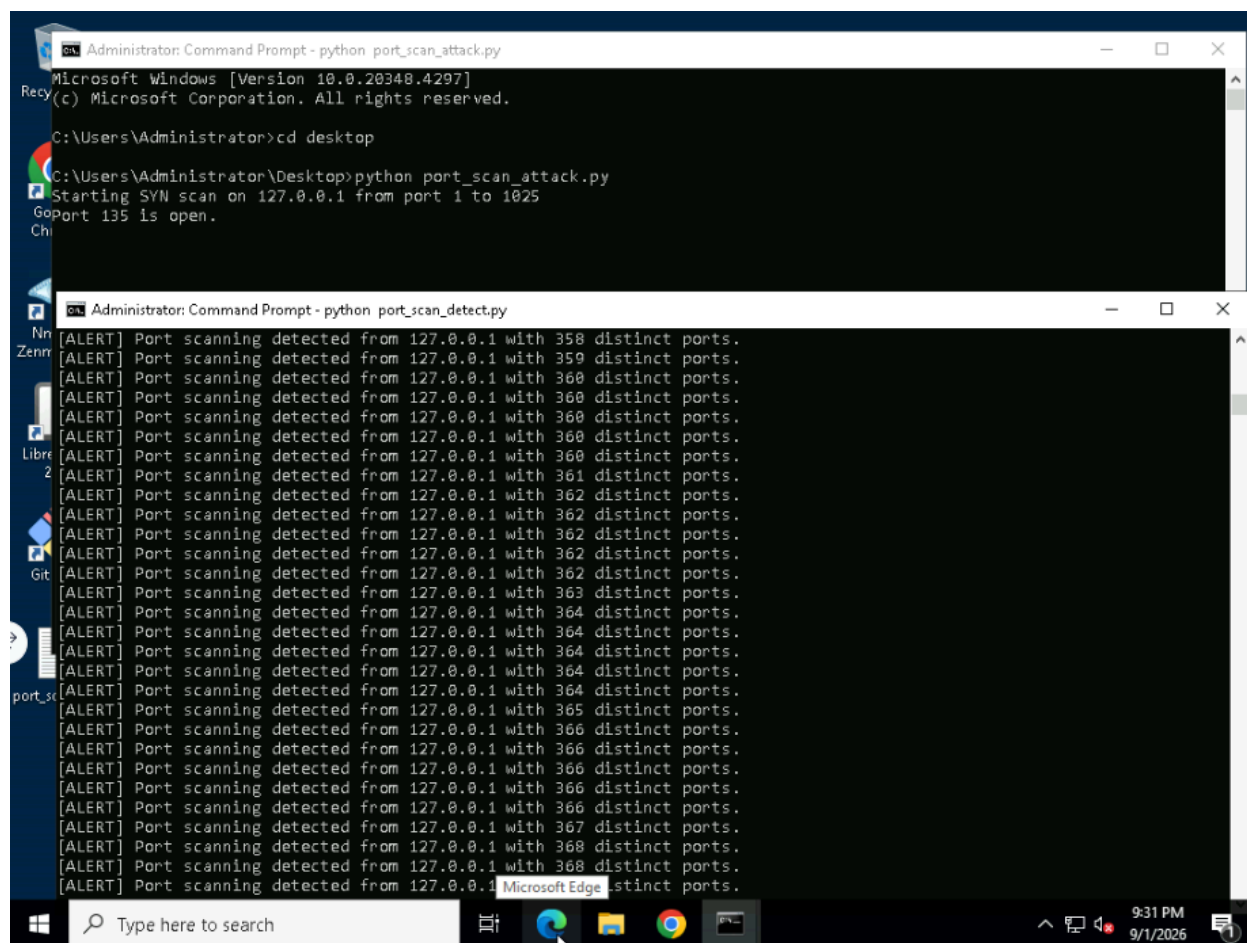


```
Administrator: Command Prompt - python port_scan_attack.py
Microsoft Windows [Version 10.0.20348.4297]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd desktop

C:\Users\Administrator\Desktop>python port_scan_attack.py
Starting SYN scan on 127.0.0.1 from port 1 to 1025
```

- 3.) Open a second command prompt window
 - a.) In command prompt navigate to the folder where port_scan_detect.py is located
 - b.) Type “cd desktop”
 - c.) Run the port scan script, type “python port_scan_detect.py”



The image shows two overlapping Windows Command Prompt windows. The top window, titled 'Administrator: Command Prompt - python port_scan_attack.py', shows the execution of a Python script. The output indicates a SYN scan on 127.0.0.1 from port 1 to 1025, with a message 'Port 135 is open.' The bottom window, titled 'Administrator: Command Prompt - python port_scan_detect.py', shows a series of 20 'ALERT' messages. Each message reports 'Port scanning detected from 127.0.0.1 with [number] distinct ports.' The number of distinct ports increases from 358 in the first message to 368 in the last message. The Windows taskbar at the bottom shows the search bar, task view button, and several application icons. The system clock indicates 9:31 PM on 9/1/2026.

```
Administrator: Command Prompt - python port_scan_attack.py
Microsoft Windows [Version 10.0.20348.4297]
(c) Microsoft Corporation. All rights reserved.

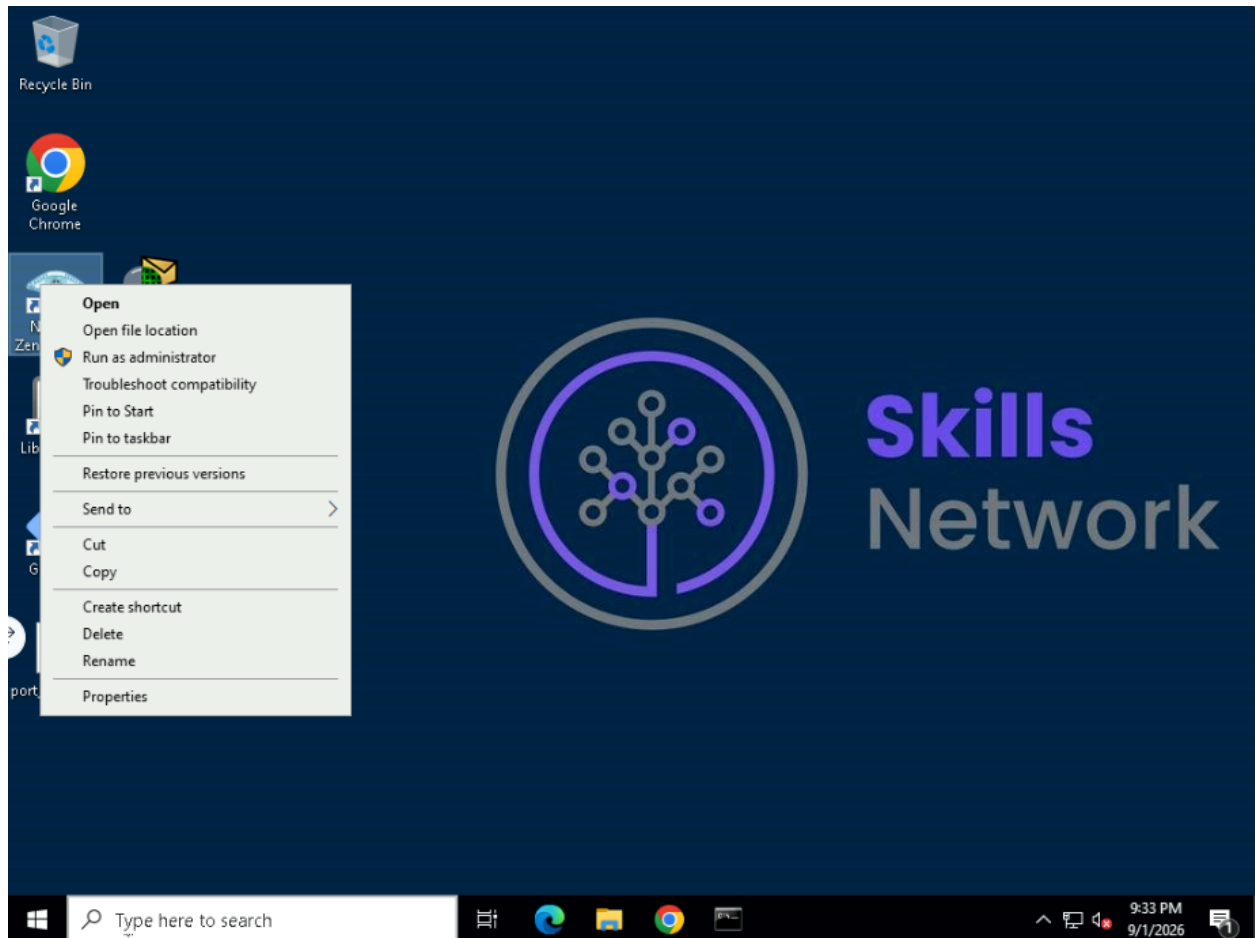
C:\Users\Administrator>cd desktop

C:\Users\Administrator\Desktop>python port_scan_attack.py
Starting SYN scan on 127.0.0.1 from port 1 to 1025
GoPort 135 is open.

Administrator: Command Prompt - python port_scan_detect.py
[ALERT] Port scanning detected from 127.0.0.1 with 358 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 359 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 360 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 360 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 360 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 360 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 361 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 362 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 362 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 362 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 362 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 363 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 364 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 364 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 364 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 364 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 365 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 366 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 366 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 366 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 366 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 367 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 368 distinct ports.
[ALERT] Port scanning detected from 127.0.0.1 with 368 distinct ports.
```

Task 6.) Using Nmap

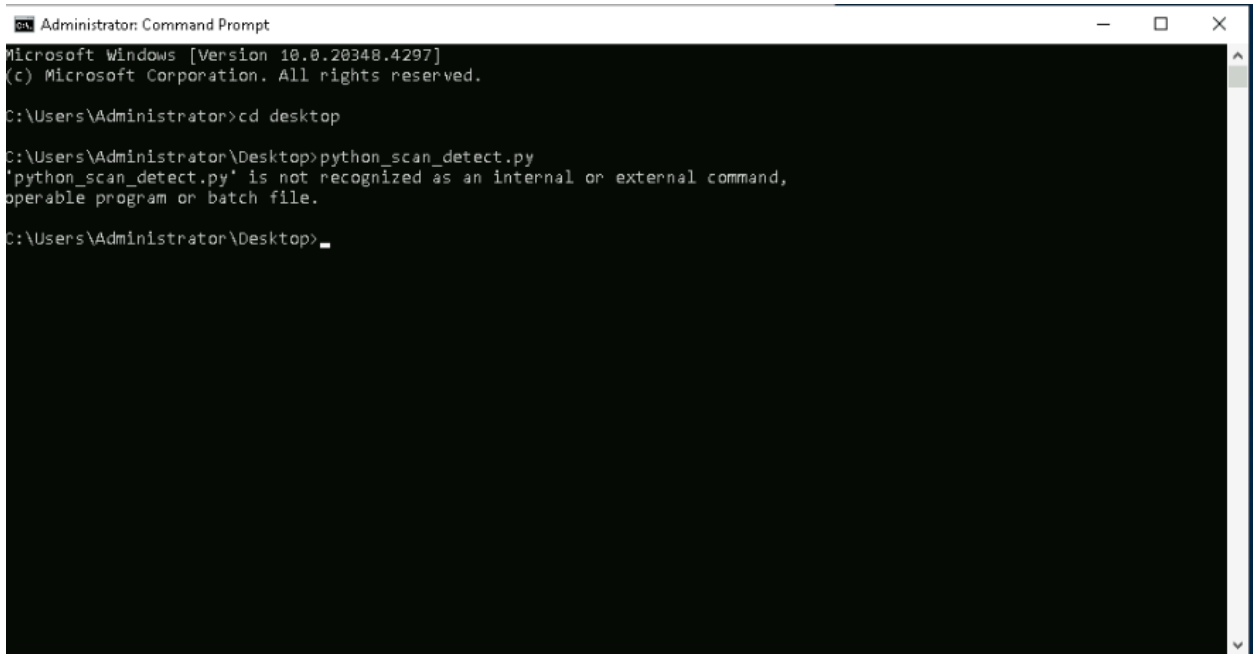
- 1.) Launch Nmap with administrative privileges



2.) Launch command prompt and select run as administrator



- 3.) Navigate to the folder where port_scan_detect.py is located
 - a.) Type “cd desktop”
 - b.) Run the port scan script with “python port_scan_detect.py”



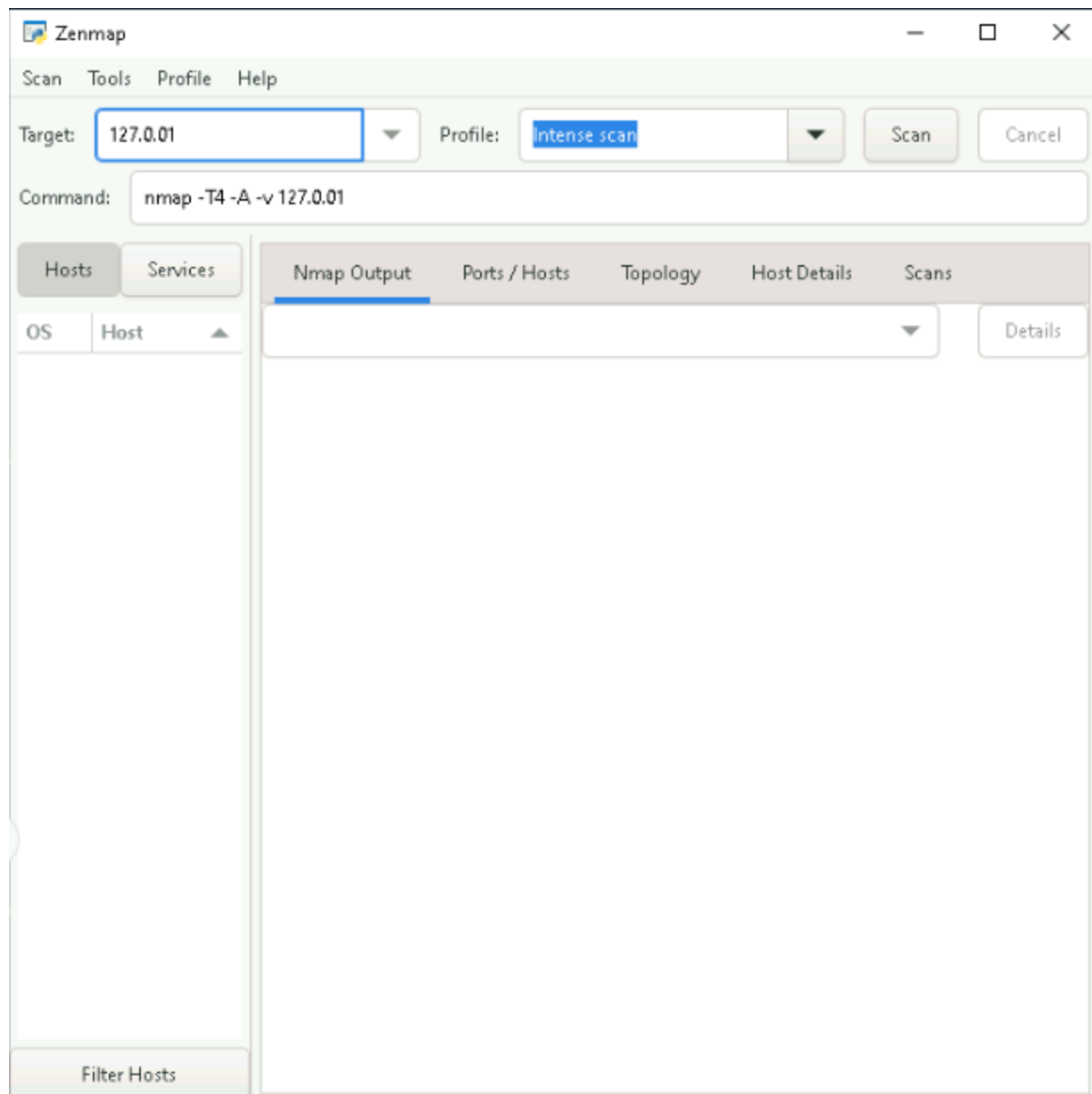
```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.4297]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd desktop

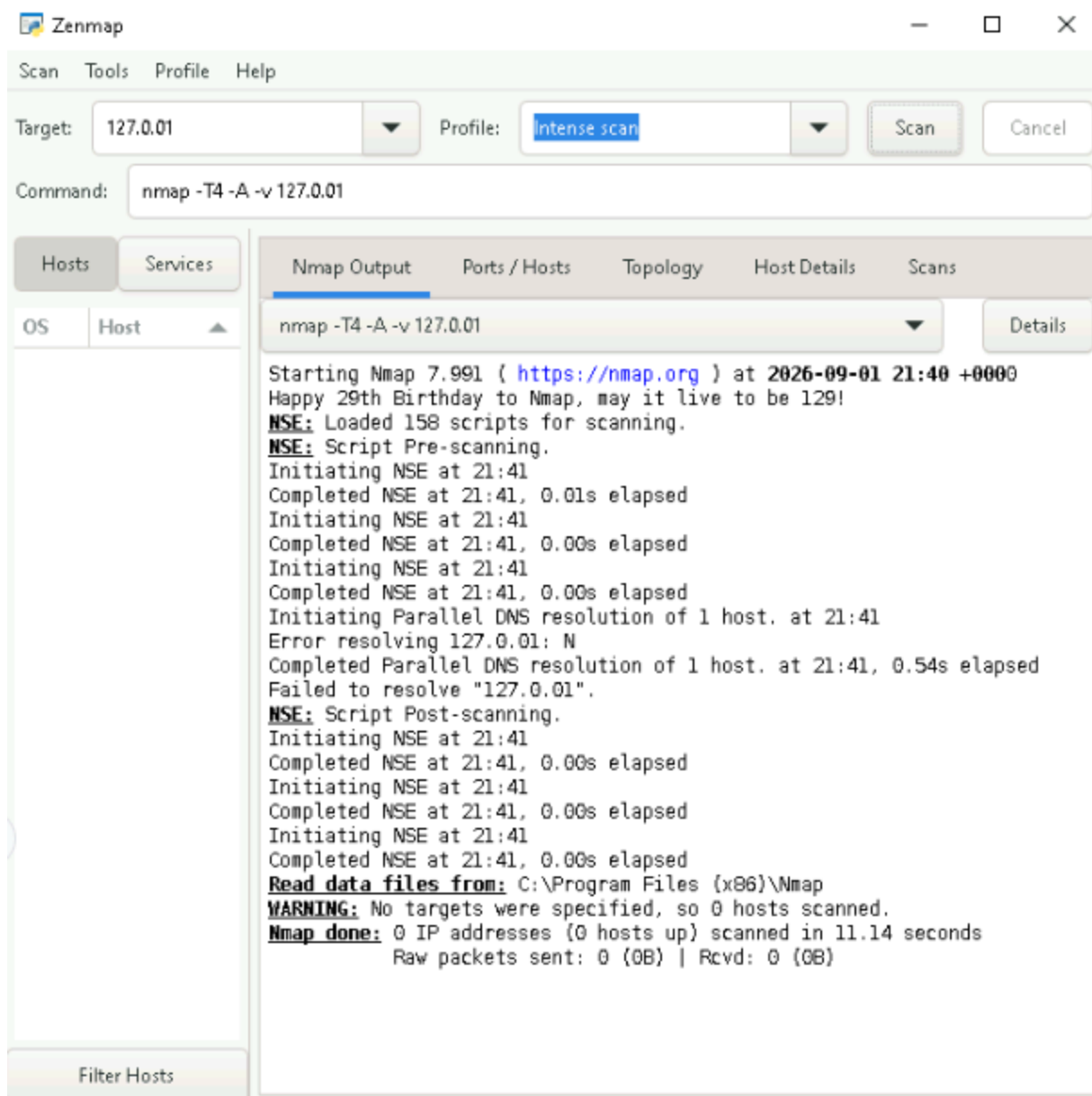
C:\Users\Administrator\Desktop>python scan_detect.py
'python_scan_detect.py' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator\Desktop>_
```

- 4.) In the Nmap interface, type the loopback IP address 127.0.0.1 in the Target field
 - a.) Select Intense Scan or Quick Scan



5.) Select Scan to begin scanning the local system



6.) Review the detection output for any alerts generated by the script